

1 Introduction to PCB

In an Operating System (OS), many programs run at the same time. Each running program is called a process.

But the computer (CPU) can execute only one process at a time. So the OS must keep track of all processes.

For this purpose, the Operating System creates a special data structure called:

👉 Process Control Block (PCB)

2 What is PCB?

Definition (Exam Definition):

A Process Control Block (PCB) is a data structure maintained by the Operating System that stores all important information about a process.

In simple words:

👉 PCB is like a file or record of a process.

👉 It contains all details needed to manage and control the process.

Without PCB, the OS cannot manage processes properly.

3 Why PCB is Important?

PCB is important because:

- It helps in process scheduling
- It helps in context switching
- It stores complete information of the process
- It allows the OS to stop and resume processes
- It helps in memory and resource management

If PCB is deleted, the process is terminated.

4 Where is PCB Stored?

PCB is stored in main memory (RAM).

The collection of all PCBs is maintained in a special area called:

👉 Process Table

Each process has its own PCB in the process table.

Main Components of PCB (Very Important for Exam)

Now we will study all fields of PCB one by one in simple language.

1. Process ID (PID)

- Each process has a unique number.
- This number is called Process ID (PID).
- It helps the OS to identify processes.

Example:

Process A → PID = 101

Process B → PID = 102

👉 Just like a roll number in class.

2. Process State

This tells the current condition of the process.

A process can be in one of these states:

1. New – Process is being created
2. Ready – Waiting for CPU
3. Running – Currently executing
4. Waiting/Blocked – Waiting for I/O
5. Terminated – Process finished

PCB stores the current state of the process.

3. Program Counter (PC)

Program Counter stores:

👉 Address of the next instruction to be executed.

When CPU stops a process and switches to another process:

- The current instruction address is saved in PCB.
- Later, the process resumes from the same point.

This is very important for context switching.

4 CPU Registers

Definition:

CPU Registers are small storage locations inside the CPU that hold temporary data and instructions during execution.

👉 PCB stores the values of all CPU registers when a process is interrupted.

👉 Simple words:

They store temporary working data of a process.

5 Priority

Definition:

Priority is a scheduling parameter that determines the importance of a process compared to other processes.



👉 Higher priority process gets CPU first.

👉 Lower priority process waits longer.

👉 Simple words:

It tells which process should run first.

Memory Management Information

Definition:

Memory Management Information includes details required for allocating and managing memory for a process.

It may include:

- Base register
- Limit register
- Page table
- Segment table

👉 Simple words:

It tells where the process is stored in memory.

I/O Status Information

Definition:

I/O Status Information contains details about input/output devices and files used by the process.

It may include:

- List of open files
- Allocated devices
- File pointers

👉 Simple words:

It tells what devices or files the process is using.

Protection Information

Definition:

Protection Information stores security-related data to protect process resources from unauthorized access.

 It includes:

- Access permissions
- Privilege level
- Security keys

 Simple words:

It ensures safety and security of the process.

Very Short Revision Version (1 Line Each)

- PID → Unique number of process
 - Process State → Current status of process
 - Program Counter → Address of next instruction
 - CPU Registers → Temporary storage of CPU
 - Priority → Importance level of process
 - Scheduling Info → Helps select next process
 - Memory Info → Memory details of process
 - I/O Info → Devices and files used
 - Accounting Info → Resource usage details
 - Protection Info → Security information
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Advantages of PCB

- ✓ Helps in multitasking
 - ✓ Makes scheduling possible
 - ✓ Helps in process management
 - ✓ Stores complete execution information
 - ✓ Supports context switching
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Disadvantages

- ✗ Takes memory space
- ✗ Large number of processes → more PCB storage

	Process ID	
	Process State	
	Program Counter	
	CPU Register	
	Memory management	
	Process Scheduling	
	I/O Information	

the process :

① New ② Ready. ③ Running ④ Waiting ⑤ Terminated.

③ Program Counter : Program Counter is a pointer which store the address of next instruction the need to be executed for the process.

④ CPU Register :

* CPU Register are small, fast storage location within a CPU

* It is used to hold data temporarily during

* It is used to hold data temporarily during processing.

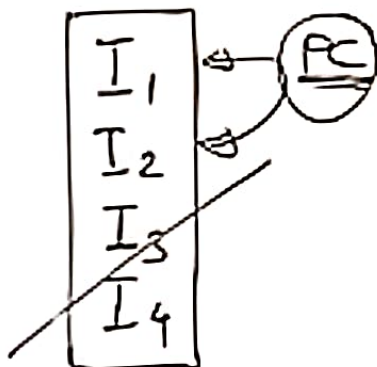
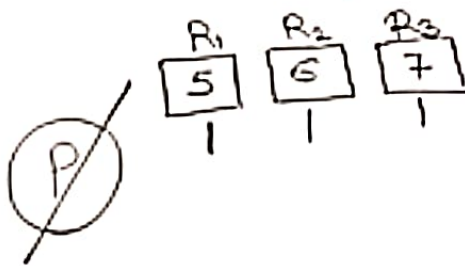
⑤ Memory Management Information:

* It store details about memory allocation

⑥ Process Scheduling Information:

→ It determines how and when Process are assigned CPU time.

Process Control Block



Process ID
Process State
Program Counter
Priority
Registers
Files List
I/O Info
Protection
etc.....

🌟 Context Switching in Operating System

(Long, detailed, very easy English notes for exam – simple and quick to remember)

1 Introduction

In a computer system, many processes run at the same time.

But the CPU can execute only one process at a time.

So the Operating System quickly switches the CPU from one process to another.

This switching process is called:



Context Switching



* Context Switching is a fundamental concept in operating system that allow the CPU to switch between different process and threads.

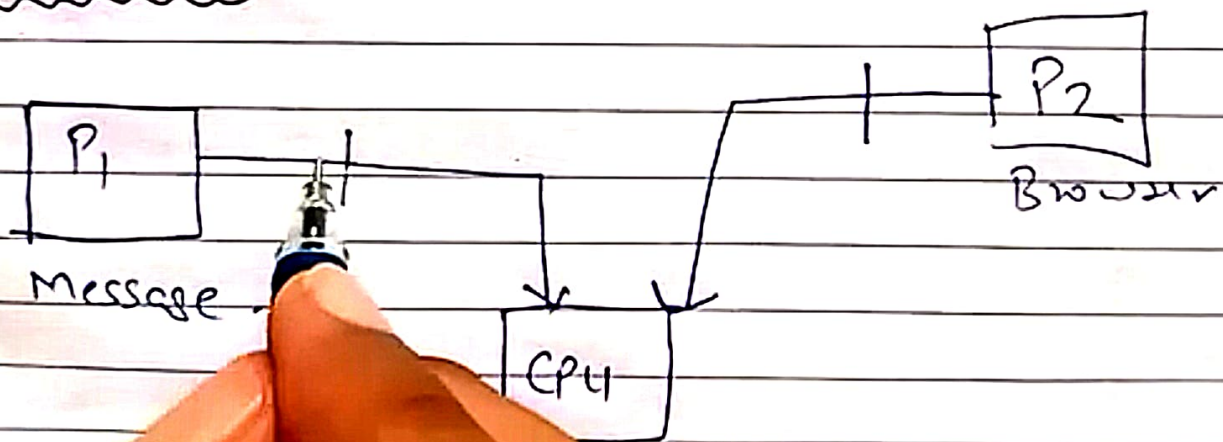
* The operating system needs to manage multiple task simultaneously, but the CPU can execute only one process at a time.

* In simple term, it allow the CPU to save the current process state, switch to different task and then later resume the original task from where it left off.

Explanation :

* In Simple term, it allow the CPU to save the current process state, Switch to different task and then later resume the original task from where it left off.

Explanation :-



What is "Context"?

The word context means the complete information about a process at a particular time.

The context of a process includes:

- Program Counter
- CPU Registers
- Process State
- Memory Information
- Stack Pointer
- Open files

All this information is stored in PCB (Process Control Block).

Why Context Switching is Needed?

Context switching is needed because:

1. CPU can run only one process at a time.
2. Many processes are waiting to execute.
3. It supports multitasking.
4. It improves CPU utilization.
5. It ensures fair scheduling.

Without context switching, multitasking is not possible.

5 Components Involved in Context Switching

Now let's understand all important components.

1 Process Control Block (PCB)

- Stores complete information of process.
 - Used to save and restore process state.
 - Very important in context switching.
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2 CPU Registers

Registers store temporary data during execution.

During switching:

- Register values are saved in PCB.
 - New process register values are loaded.
-

3 Program Counter (PC)

- Stores address of next instruction.
- Saved before switching.
- Restored when process resumes.

◆ What is Stack Pointer?

Definition (Exam Ready):

Stack Pointer is a special register that stores the address of the top element of the stack.

👉 Simple words:

Stack Pointer tells where the top of the stack is.



Scheduler

Now let's understand Scheduler.

♦ What is Scheduler?

Definition (Exam Ready):

Scheduler is a part of the Operating System that selects which process will run next.



Simple words:

Scheduler decides which process gets the CPU.

♦ Types of Scheduler (Important)

1 Long-Term Scheduler

- Decides which process enters the system.
- Runs less frequently.

2 Short-Term Scheduler (Most Important)

- Selects process from ready queue.
- Runs very frequently.

3 Medium-Term Scheduler

- Suspends or resumes processes.

Dispatcher

Now the most confusing one 😊

- ◆ What is Dispatcher?

Definition (Exam Ready):

Dispatcher is a module of the Operating System that gives control of the CPU to the process selected by the scheduler.

👉 Simple words:

Scheduler selects

Dispatcher actually gives CPU

6 How Context Switching Works (Step-by-Step)

Let's understand with two processes:

Process P1

Process P2

Step 1: P1 is Running

- CPU is executing P1.
 - Registers contain P1 data.
 - Program Counter points to next instruction of P1.
-

Step 2: Interrupt Occurs

Interrupt may occur due to:

- Time slice complet 
 - I/O request
 - Higher priority process arrived
-

Step 3: Save P1 State

Operating System:

- Saves Program Counter of P1
- Saves CPU registers
- Saves stack pointer
- Updates process state

All saved into P1's PCB.

Step 4: Select P2

Scheduler selects P2 from ready queue.

Step 5: Restore P2 State

OS:

- Loads P2's register values
- Loads P2's Program Counter
- Restores stack pointer
- Changes process state to Running

Now P2 starts executing.

Step 6: Repeat

When P2 stops, same process happens again.

7 Diagram Explanation (Based on Your Image)

From your image:

- P1 is executing instructions (I1, I2, I3, I4)
- CPU registers contain some values (like 3, 4, 5)
- Suddenly switching happens
- P1 state is saved
- P2 state is restored
- CPU now executes P2 instructions

Right side shows:

Save P1 State

Restore P2 State

Save P2 State

Restore P1 State

This continuous switching is called Context Switching.

Q3) How Context Switching Works?

SUBSCRIBE
NOW

- ① The Current process is Interrupted, either due to an external event (like I/O) or because its time slice over.
- ② The operating System Save the Current State of the process in its PCB.
- ③ The OS then loads the Saved of another process from its PCB.
- ④ The new process starts execution.

- ① The Current process is Interrupted, either due to an external event (like I/O) or because its time slice over.
- ② The operating System Save the Current State of the process in its PCB.
- ③ The OS then loads the Saved of another process from its PCB.
- ④ The new process now take Control of CPU and start executing.

8 Types of Context Switching

1 Voluntary Context Switching

Happens when process:

- Requests I/O
- Waits for resource

Process willingly gives up CPU.

2 Involuntary Context Switching

Happens when:

- Time slice ends
- Higher priority process arrives

OS forces process to stop.

Advantages of Context Switching

- ✓ Supports multitasking
 - ✓ Improves CPU utilization
 - ✓ Ensures fairness
 - ✓ Handles priority scheduling
-

Disadvantages

- ✗ Wastes CPU time
 - ✗ Increases overhead
 - ✗ Too many switches reduce performance
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Why PCB is Important in Scheduling?

Without PCB:

- ✗ OS cannot know process state
- ✗ OS cannot save registers
- ✗ Scheduler cannot choose process
- ✗ Context switching cannot happen

So PCB is the backbone of CPU scheduling.

5 Marks Answer (Ready to Write)

PCB stores important information about a process such as Process ID, Process State, Program Counter, CPU registers, Priority, Memory information, I/O information and Scheduling information. In CPU scheduling, PCB plays an important role because the scheduler uses the information stored in PCB to select the next process. During context switching, the state of the current process is saved in its PCB and the state of the next process is restored from its PCB. Therefore, PCB is essential for process management and scheduling.

Context Switching

